

WIRELESS COMMUNICATION · SESSION PREP

5G & the Road to 6G

Presenter preparation guide — slide-by-slide content & talk track

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26 slides · 4 parts + open floor · Companion to [5g-to-6g-session.html](#)

How to use this guide

- **One block per slide.** Each block shows the slide number/title, **“On screen”** (what the audience sees) and **“Talk track”** (what you say — expand naturally, don’t read it).
- **The anchor slide is #5 “One signal, five generations.”** Everything else hangs off the idea that each generation solved the previous one’s hard limit. If you’re short on time, protect that slide.
- **Lean into your edge.** You’re a hardware engineer — slides on Massive MIMO/beamforming (#11), the three domains (#13), and roles (#23) are where you can go off-script with real experience from Cavli.
- **Keep the big numbers honest.** 20 Gbps, 1 ms, ~1 Tbps are *ITU/research targets*, not what a phone shows. Saying so builds credibility.
- **Timing:** ~25–30 min of content leaves room for Q&A. Part openers (the “Part One/Two…” slides) are 10-second breathers — use them to reset pace.
- **Audience hooks:** show of hands on slide #7 (3G download wait); invite their field on slides #14, #19.

5G & the Road to 6G

ON SCREEN

- Title with animated signal waveform.
- Tagline: “From today’s intelligent networks to the intelligence-native networks ahead.”
- Your name & role.

TALK TRACK

Welcome. Introduce yourself — Irene Binnet, Hardware Design Engineer at Cavli Wireless. Set the agenda in one breath: how mobile networks evolved into 5G, where 6G is heading, and the careers behind all of it. Invite questions throughout, with an open floor at the end.

What we’ll cover

ON SCREEN

- **Evolution of mobile networks** — analog voice to an intelligent ecosystem, and what drove each leap.
- **Understanding 5G** — core tech, service classes, applications, deployment realities.
- **The road to 6G** — why it’s coming, enabling tech, experiences unlocked.
- **Careers in telecom & networking** — roles, emerging opportunities, skills.

TALK TRACK

Four areas, then open Q&A: from history, to how 5G works, to the 6G outlook, to where the careers are. Keep it to ~20 seconds — it’s a signpost, not a topic.

Evolution of Mobile Networks

ON SCREEN

- Section opener. Thesis: “Every generation existed to solve the hard limit of the one before it.”

TALK TRACK

Plant the thread that carries the whole talk: each generation solved a specific limitation of the last. Say it explicitly here so you can call back to it on every later slide.

Connectivity is infrastructure now

ON SCREEN

- From signals and letters to **instant global communication** in a generation.
- Mobile put the internet in everyone's pocket and **reshaped how society runs**.
- An outage halts business and services — it's **critical infrastructure**, not a convenience.
- Where it lands: business, healthcare, education, entertainment.

TALK TRACK

Frame connectivity as infrastructure, not a gadget. Tie the sector examples to whatever this room works with — pick the one or two closest to the audience and make it concrete.

One signal, five generations

ON SCREEN

- **1G** (1980s) — Analog voice. First mobile calls, no data.
- **2G** (1990s) — Digital + SMS. Texting, clearer calls.
- **3G** (2000s) — Mobile internet. Web & multimedia.
- **4G** (2010s) — LTE broadband. HD video streaming.
- **5G** (2020s) — Connected ecosystem. Low latency, massive scale.

TALK TRACK

The deck's anchor — slow down here. Walk left to right: voice, text, internet, video, machines. Point at the waveforms getting denser across the row — each generation carries more, and a *new kind* of information, not just more speed. Every later slide refers back to this picture.

If short on time: protect this slide. Cut elsewhere first.

Why networks never stand still

ON SCREEN

- **Growing data consumption** — video, cloud, high-res content raise the bandwidth floor.
- **Explosion of devices** — phones, wearables, sensors, machines (IoT at scale).
- **Real-time expectations** — gaming, calls, AR/VR, control systems.
- **Automation & M2M** — machines talking to machines.

TALK TRACK

These four forces compound, so demand never settles — that's *why* there's always a next generation. Drop the stat: connected devices already outnumber people.

What the leaps actually feel like

ON SCREEN — DOWNLOAD A 2-HOUR MOVIE (~5 GB)

NETWORK	TIME	THROUGHPUT
3G	2–3 hours	~2–5 Mbps
4G	15–30 minutes	~20–50 Mbps
5G	Seconds to a minute	hundreds of Mbps to Gbps

TALK TRACK

A benchmark people *feel*. Quick show of hands — who remembers waiting on a 3G download? Bridge out of it: expectations now assume instant, and that pressure is exactly what drives the next generation.

Understanding 5G

ON SCREEN

- Section opener. Thesis: “Not a faster 4G — a platform built for people, machines, and mission-critical control at once.”

TALK TRACK

The core of the talk — signal you’re going under the hood now: what makes 5G different, then where it’s used.

What is 5G?

ON SCREEN

- Fifth generation of mobile networks — an **intelligent, connected ecosystem** for people, machines, and industries.
- Objectives: extreme speed & capacity · ultra-low latency · massive device connectivity · mission-critical reliability.

20 Gbps peak data-rate target (ITU IMT-2020)

~1 ms air-interface latency target

10⁶ / km² connected-device density

TALK TRACK

Stress that 5G is a *platform*, not a faster phone. The three numbers are ITU IMT-2020 targets — ideal, not what a handset shows today. Set that expectation honestly; it makes you more credible, not less.

4G versus 5G

CAPABILITY	4G LTE	5G
Peak data rate	~1 Gbps	up to ~20 Gbps
Typical latency	~30–50 ms	~1–10 ms
Device density	~10 ⁵ / km ²	~10 ⁶ / km ²
Mobility	up to ~350 km/h	up to ~500 km/h
Primary focus	Mobile broadband	Broadband + IoT + critical control

TALK TRACK

Walk the rows. The real game-changers for industry are **latency and density**, not raw peak speed — that's what enables control loops, automation, and dense IoT. Note these are targets/typical values; real figures vary by deployment and spectrum.

Massive MIMO & beamforming

ON SCREEN

- **Massive MIMO** (Multiple-Input, Multiple-Output) — dozens-to-hundreds of antennas per cell, serving many users at once, far more capacity & coverage.
- **Beamforming** — focused, steerable signals aimed at each user; less waste, better signal strength & efficiency.

TALK TRACK

MIMO = more antennas, more parallel streams. Beamforming = aiming instead of broadcasting — **floodlight vs. steerable spotlight**. This is your world as a hardware/RF engineer — go off-script here with a real example.

Your edge: spend extra time. This is the most hardware-native slide in the deck.

An architecture defined by 3GPP

ON SCREEN

- 3GPP (3rd Generation Partnership Project) — the global body that standardizes mobile networks.
- Chain: **UE** → **NG-RAN** → **5G Core** → **Data Network**, with reference points Uu (air), N2/N3, N6.
- Release strip: Rel-15 (first 5G, 2018) · NR (new radio) · 5GC · NSA → SA.

TALK TRACK

5G isn't one vendor's design — it's a shared specification, published in numbered Releases (Rel-15 first full 5G in 2018; Rel-16/17/18 add "5G-Advanced"). The arrow labels (Uu, N2/N3, N6) are the actual 3GPP reference points. NSA reused the 4G core first; SA is the full 5G core. Walk it left to right.

UE, RAN & 5G Core

ON SCREEN

- **UE (User Equipment)** — device + 5G modem + (U)SIM; phones, vehicles, cellular IoT modules; speaks NR over the air.
- **NG-RAN** — the radio layer; gNB base stations with Massive MIMO/beamforming; split into CU / DU / RU.
- **5G Core (5GC)** — cloud-native, service-based; functions AMF / SMF / UPF; control-plane / user-plane split.

TALK TRACK

UE is your world — the device, modem, and SIM; at Cavli you build cellular IoT modules, which *are* UE. RAN is the radio: gNB, now split into CU/DU/RU for flexible deployment. The Core is cloud-native and service-based — AMF (access & mobility), SMF (sessions), UPF (moves user traffic). The control/user-plane split is what makes network slicing and edge breakout possible.

Where 5G shows up

ON SCREEN

- Smart factories · autonomous vehicles · remote surgery · smart cities · AR/VR · drones, logistics, agriculture.

TALK TRACK

Don't read every tile — pick two or three closest to the audience and go deep. You can speak to the device/module side of connecting any of these. Good moment to ask what the room would use it for.

What still has to be solved

ON SCREEN

- Challenges: **infrastructure cost** · **spectrum management** · **security** (larger attack surface) · **rural coverage**.
- Already in production: **Private 5G** in manufacturing · **5G healthcare** (diagnostics, remote monitoring, telemedicine).

TALK TRACK

Be honest about what's hard — cost, spectrum, security, rural reach. Then balance it: Private 5G is a real, growing commercial segment and a concrete example of where investment is flowing.

The Road to 6G

ON SCREEN

- Section opener. Thesis: “Still research today — but the direction is set: faster, near-instant, and intelligence-native.”

TALK TRACK

Shift to looking ahead. Flag clearly that 6G is research today — but the direction is clear, and it’s where a lot of current work points.

17 6G • THE NEXT LEAP

Why 6G, and what to expect

ON SCREEN

- Why: today’s networks cap out for the most demanding future apps; relentless data growth; immersive, AI-driven experiences 5G can’t fully serve.
- Expected: **~1 Tbps** (research target) · **near-zero / sub-ms latency** · **AI-native** · enhanced end-to-end reliability.

TALK TRACK

The headline: 6G isn’t just faster — it’s **AI-native** with near-zero latency. Frame ~1 Tbps as an aspirational research target, not a committed spec, to keep your numbers credible.

18 6G • ENABLING TECHNOLOGIES

What makes 6G possible

ON SCREEN

- **AI in the network** — self-optimizing, predictive maintenance, autonomous management.
- **Terahertz (THz) communication** — very-high-frequency spectrum for ultra-high data rates.
- **Space–air–ground networks** — satellites + drones + terrestrial, integrated for global reach.
- **Digital twins** — live virtual replicas of physical systems.

TALK TRACK

Four enablers. Terahertz and space–air–ground are the most physically novel — and the hardest hardware problems (your wheelhouse). Digital twins are a bridge to industry the audience may already know.

The experiences it unlocks

ON SCREEN

- Holographic comms · extended reality · brain–computer interfaces · advanced robotics · autonomous transport · the unimagined.

TALK TRACK

Directional, not promises — keep the tone exploratory. Good moment to invite the room: which of these would matter in *their* field?

Research & standardization

ON SCREEN

- Active global research across academia, industry, and standards bodies (ITU, 3GPP).
- 6G is being **defined now** — spectrum, requirements, use cases.
- Today's engineers are the ones who will build it.

~2030 expected first commercialization

NOW research & standardization underway

TALK TRACK

We're in the standardization phase; ITU/3GPP timelines point to roughly 2030. Land the opportunity: anyone entering the field now is early enough to shape it.

Careers in Telecom & Networking

ON SCREEN

- Section opener. Thesis: “Everything we just covered is built by people — here's where they sit, and how to get there.”

TALK TRACK

The practical part — where the roles are and how to move into them. Speak from your own experience here.

Why this field

ON SCREEN

- **Foundation of everything digital** — every app, cloud service, and connected device (including AI) rides on these networks.
- **Continuous reinvention** — 5G, 6G, AI, satellites; renews every few years.
- **Durable global demand** — skilled engineers needed worldwide, across every industry.

TALK TRACK

Position telecom/networking as foundational and durable. Make the point that every digital trend — AI and cloud included — rides on this infrastructure, so the demand isn't a fad.

Telecommunications roles

ON SCREEN

- RF Engineer · Wireless Comm. Engineer · Core Network Engineer · GNSS & Location · Satellite Comm. Engineer · **Hardware Design** (your role at Cavli).

TALK TRACK

Concrete roles. You sit on the hardware/RF side — offer to share what the day-to-day actually looks like at a cellular module company like Cavli Wireless. This personal angle is what they'll remember.

Networking & emerging roles

ON SCREEN

- **Networking:** Network Engineer · Network Security · Cloud Networking · Data Center Engineer.
- **Emerging:** 5G/6G Research · AI for Networks · Private Network · Edge Computing · Network Automation · Cybersecurity Analyst.

TALK TRACK

Classic networking roles plus the newer titles from 5G/6G, automation, and AI. Call out that edge computing and cybersecurity are hiring heavily right now.

Skills that get you in

ON SCREEN

- **Technical:** networking fundamentals & TCP/IP · Linux · wireless communication · Python scripting · cloud.
- **Human:** problem-solving · communication · collaboration · continuous learning.

TALK TRACK

The actionable takeaway — make it land. Fundamentals + Linux + Python are the common base. Push **hands-on projects and labs over certificates alone.**

Let's talk.

ON SCREEN

- Closing. “Questions, perspectives, and where the industry goes next.” Your name & role again.

TALK TRACK

Open the floor. Keep a couple of seed questions ready (e.g. “Where would you want to specialize?”). Thank everyone and share contact details if appropriate.

Seed questions to have ready: What surprised you most about 5G vs 6G? · Which application matters most in your field? · Hardware vs. software side — which pulls you?
